Neil Janwani

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EDUCATION

PH.D. **Robotics**, Georgia Institute of Technology — Aug. 2024 — June 2029
Research Advisor: Prof. Maegan Tucker (Dynamic Mobility Lab). — GPA: 4.0

B.S. **Computer Science**, California Institute of Technology — Sep. 2020 — June 2024
Research Advisors: Prof. Aaron Ames, Prof. Joel Burdick. — GPA: 4.0

RESEARCH HIGHLIGHTS: *Whole-body control and personalization of humanoid and assistive wearable robots.*

Robust Humanoid Motion Imitation and Whole-Body Control. Jan. 2025 — Present
Encoding when and how humanoid controllers should deviate from idealized target motions improves robustness.

- Improved imitation accuracy and adherence to gait style via an RL policy that selects which reference to track [4].
- Built a continuous gait library by smoothly interpolating between dynamically feasible reference motions produced via trajectory optimization. Achieved hardware deployment on the BRUCE humanoid robot [4].
- Designed a residual RL framework to tune the trade-off between imitation fidelity and robustness in humanoid robots.
- Applied this method to train motion-tracking policies on LAFAN1 human motion-capture data. Achieved zero-shot sim-to-real transfer on the Unitree G1.

Multi-Objective Robotic Optimization. June 2025 — Present
Discovering Pareto sets of controllers over several objectives enables high-quality, diverse personalization at deployment time.

- Learned Pareto-optimal whole-body motion imitation across 6 objectives in $\sim$2 hrs on the BRUCE humanoid [3].
- Achieved 270$\times$ speed-up of Pareto set discovery via GPU-acceleration of hypernetwork-based multi-objective PPO [3].
- Devised a multi-objective co-design framework, discovering a specialized robot design and control for each trade-off [2].

Human-Robot Interaction and Personalization. June 2021 — Present
Robotic personalization and aligned human-robot teams can improve robot integration in the real world.

- Created a multi-objective Bayesian optimization framework for sample-efficient, personalized Pareto set discovery [1].
- Designed and ran a human-subject study validating this framework on a lower-limb exoskeleton [1].
- Improved operator-robot alignment without losing formal safety guarantees with an intention-aware CBF [5].
- Designed motor control and gait-phase inference of a soft ankle exoskeleton and ran human-subject experiments [6].

RESEARCH & WORK EXPERIENCE

Dynamic Mobility Lab, Georgia Tech — Atlanta, Georgia
NSF and Georgia Tech IRIM Graduate Research Fellow — Aug. 2024 — Present

- Researching whole-body control and personalization of humanoid and assistive robots detailed above.

Burdick Group, Caltech — Pasadena, California
Research Fellow: Kiyo and Eiko Tomiyasu Named Scholar — June 2023 — June 2024

- Devised a CBF safety filter that preserves formal safety guarantees while allowing operator-specific personalization [5].
- Developed a haptic GUI communicating live robot safety data for Caltech's LINC DARPA team.

MIT Lincoln Laboratory — Lexington, Massachusetts
Research Intern — June 2022 — Sep. 2022

- Built the control stack of sensorless gimbal motor, estimating steady-state electrical phase in under 20 sec.

AMBER Lab, Caltech — Pasadena, California
Research Fellow — June 2021 — June 2024

- Trained an LSTM gait-phase approximator from data collected via ankle-mounted IMUs for stride-based control [6].
- Designed and built a custom ankle exoskeleton, including PCB design, biometric sensor integration, FOC-based motor control, and safety-critical hardware constraints for wearable use [6].

TECHNICAL SKILLS

- **Programming Languages:** `Python, C/C++`
- **Software:** JAX, PyTorch, BoTorch, OpenCV, CVXPY, MuJoCo, Gymnasium, mjlab, MJX, Isaac Lab, ROS1&2, Gazebo, Linux, Git, CMake, MATLAB, Mathematica, SolidWorks, Weights & Biases, Claude Code, GitHub Copilot
- **Hardware:** Unitree G1 Humanoid, Westwood Robotics BRUCE Humanoid, custom (Google X) lower-limb exoskeleton, Wandercraft Atalante Exoskeleton, (custom) ankle exoskeleton, DEVCOM GVR-BOT, 6-DOF Robotic Manipulators

PUBLICATIONS

In review:

[1] **Janwani N.**, Lerner M. T., Young A. J., Tucker M. "Multi-Objective Human-in-the-Loop Bayesian Optimization of a Lower-Limb Exoskeleton." *In review for ICRA 2027* [pdf][website]

[2] Madabushi V., **Janwani N.**, Tucker M. "MOCHA: Multi-Objective Co-Design using Hypernetwork Architectures." *In review for ICRA 2027* [pdf]

Journal Publications:

[3] **Janwani N.**, Novoseller E., Lawhern V., Tucker M. "MO-Playground: Massively Parallelized Multi-Objective Reinforcement Learning for Robotics." *IEEE Robotics and Automation Letters, 2026* [Webpage]

Conference Publications:

[4] **Janwani N.**†, Madabushi V.†, Tucker M. "NaviGait: Navigating Dynamically Feasible Gait Libraries using Deep Reinforcement Learning." *IEEE International Conference on Robotics and Automation (ICRA), 2026* [Webpage]

[5] **Janwani N.**†, Daş E†., Touma T., Wei S. X., Molnar T. G., & Burdick J. W. "A Learning-Based Framework for Safe Human-Robot Collaboration with Multiple Backup Control Barrier Functions." *IEEE International Conference on Robotics and Automation (ICRA), 2024* [Paper] [Video]

Patents:

[6] Shaikewitz L., Tucker M., **Janwani N.**, Ames A. "Front-leg assistive exoskeleton" *US20240245569A1* [Patent]

SELECTED COURSE PROJECTS

- **Socially Reasonable Humanoid Gestures:** Retargeted human social gestures from monocular video to the Unitree G1 via pose landmarks and end-effector path-error minimization; sequenced motions with an LLM social task planner.
- **Learning Dynamic Motion Priors Online:** Built an agentic framework using VLMs to generate high-level task plans from natural language goals, and efficiently request one-shot demonstrations when the robot's skill library is insufficient.
- **Human-Robot Manipulation:** 5-DOF manipulator to play *Carrom* in a dynamic environment with human opponents.
- **Exploration-Based Planner:** Information-based exploration of an unknown space with probabilistic sensors.
- **Graph Search for Replanning:** Compared efficiency of D* to A* and Dijkstra for replanning on large graphs.
- **Quadruped Kinematics:** Wrote forward/inverse kinematics to simulate a quadrupedal robot designed in Solidworks.
- **Hierarchical Quadruped Planner:** Wrote a hierarchical A* planner for quadrupedal locomotion on diverse terrains.
- **Mobile Robotics:** Implemented localization, obstacle avoidance, and odometry algorithms on hardware.

TEACHING EXPERIENCE

Caltech Robotics and Computer Science Pasadena, California
Course Founder and Teaching Assistant Jan. 2022 — June 2024

- Developed and taught a 10-week, 50-student course on prototyping robotic and computing systems.
- Awarded $3300 (Housner Fund) from Caltech in funding for a larger iteration of the course for the 2024 cycle.
- Designed a mobile robot as Robotics Head Teaching Assistant for a 35-student diversity, equity and inclusion program.
- Held office hours and wrote exam materials for classes of 100+ students in Experimental Robotics (ROS2: Python), Kinematics (ROS2: Python), Data Structures (Java), Software Design (C), and Computing Systems (x86 Assembly).

AWARDS

NSF Graduate Research Fellowship Apr. 2024
Fellowship awarded by the National Science Foundation (NSF). Atlanta, Georgia

IRIM Graduate Research Fellowship Apr. 2024
Fellowship awarded by Georgia Tech's Institute of Robotics and Intelligent Machines (IRIM). Atlanta, Georgia

Thomas A. Tisch Prize for Undergraduate Teaching in Computing and Mathematical Sciences Apr. 2024
Awarded for development and instruction of a new robotics course at Caltech. Pasadena, California

Housner Fund Dec. 2023
Granted $3300 to develop a Caltech robotics course, supporting increased student interest. Pasadena, California

Kiyo and Eiko Tomiyasu Research Fellowship July 2023
Awarded a fully funded summer undergraduate research position at the Burdick Group at Caltech. Pasadena, California